GOKUL MALLABATHULA

CS UNDERGRADUATE · AI / ML

📞 6303246108
✉ Gokulmallabathula@gmail.com
🔗 www.linkedin.com/in/gokul-mallabathula-693a74118
📍 Vijayawada, India

ABOUT ME

Computer Science undergraduate at VIT-AP University with strong foundations in Python, software development, and data analysis. Experienced in building health-tech and ML applications using Flask, React, Next.js, and Supabase. Completed professional internships as a Fraud Detection Engineer and ML Engineer. Passionate about AI, data science, and developing scalable software solutions with real-world impact.

EXPERIENCE

FRAUD DETECTION ENGINEER INTERN

Jan 2026 – Mar 2026 – *Zetheta Algorithms Pvt. Ltd.*

- Worked on multiple proprietary fraud detection projects for financial security systems.
- Gained exposure to real-world anomaly detection and financial data pipelines.

MACHINE LEARNING ENGINEER INTERN

July 2026 – Aug 2026 – *FlyRank AI*

- Working as a Machine Learning Engineer Intern applying ML to search and analytics data, focused on embeddings, clustering, and intent modeling to improve product-facing AI features.
- Contributing to production-level ML pipelines using Python, pandas, and scikit-learn, applying model optimization and data-driven techniques to solve real-world search and analytics challenges.

PROJECTS

SAFEPULSE | CARDIOVASCULAR RISK ANALYSIS

Python · TensorFlow · Scikit-learn · XGBoost · SHAP · Vercel · safepulse.in

- Built classification models (Logistic Regression, Random Forest, XGBoost) to predict cardiovascular disease risk from clinical data with end-to-end ML pipeline.
- Implemented missing value imputation, feature scaling, class imbalance handling, cross-validation, and SHAP explainability; deployed with custom DNS.

RURALLINK AI | MULTILINGUAL RURAL SERVICE DISCOVERY

Next.js · TypeScript · Supabase · Gemini 1.5 Flash · Web Speech API · Mapbox API · Vercel

- Built a multilingual voice-based web app helping rural Indian users find public services, using Gemini AI for natural language understanding and Web Speech API for voice input, and Mapbox API for location-based service mapping.
- Developed the AI pipeline and API routes (/api/query, /api/journal, /api/feedback); advanced to the Final Round of USAII Global AI Hackathon 2026.

CARE TRANSITION EFFICIENCY ANALYTICS

Python · Pandas · Streamlit · Plotly · Machine Learning

- Engineered 5 process efficiency KPIs from 720 daily HHS UAC Program records (2023–2025), modelling the CBP → HHS → Sponsor care pipeline to detect bottlenecks.
- Built an interactive Streamlit dashboard with real-time KPI filtering and automated bottleneck alerts; published findings in a full academic research paper.

ParkSmart AI | Smart Vehicle Parking Management System

TypeScript, ReactJS, Vite, Supabase, SQL

- Developed an AI-powered web application for vehicle entry/exit management, parking slot allocation, and real-time parking availability tracking.
- Implemented secure authentication, database operations, and intelligent parking recommendations, improving parking management efficiency and user experience.

ARCHLENS AI | AI-POWERED CODE & ARCHITECTURE INTELLIGENCE

FastAPI · Python · NetworkX · Pydantic · OpenAI · AST · LLMs

🚧 Under Construction

- Building an AI-powered code intelligence platform that analyzes codebases, generates architecture and dependency graphs, and identifies structural relationships across components.
- Implementing architecture health analysis, risk detection, evolution tracking, refactoring insights, and grounded LLM-powered explainations.

DEEPFAKE SHIELD | MULTIMODAL AI DIGITAL MEDIA FORENSICS

PyTorch · EfficientNet · Transformers · FastAPI · Next.js · React · TypeScript

🚧 Under Construction

- Building a multimodal AI platform for detecting and analysing **deepfake video, image, and audio content** using deep learning, temporal analysis, and explainable AI.
- Developing the ML inference pipeline and FastAPI backend, integrating trained models with a Next.js frontend for real-time detection, localization, and explainable forensic results..

EDUCATION

2024 – Present

B.Tech – Computer Science & Engineering
VIT-AP University

Class XII (AP Board)
89%

Class X (CBSE)
91.6%

CERTIFICATIONS

Generative AI Workshop
IIT Hyderabad
LLMs, Prompt Engineering, AI Applications

Designing Scalable RAG Pipelines
Scaler Masterclass
RAG, Scalable Ai Pipelines, Vector Databases

PROFILES

GitHub
Portfolio

SKILLS

Languages: Python, Java

Web: React, TypeScript, Next.js, Vite, Supabase

Databases: SQL, SQLite, PostgreSQL

ML / DS: Pandas, NumPy, EDA, Scikit-learn, TensorFlow, SHAP

Tools: Git, GitHub, Jupyter, Vercel, Flask, FastAPI, Docker

LANGUAGES

English · Telugu · Hindi

ACHIEVEMENTS & ACTIVITIES

— Finalist — USAII Global AI Hackathon 2026 (Final Round) · AI/Full-Stack Developer & Lead · Team Byte Coders

— Round 2 Qualifier — Adobe University Hackathon 2026 · AI / Agents & Software Development · Team of 2–3

— Shortlisted — Smart India Hackathon 2025 (Internal Round) · Frontend Developer · Team Shark

— Participated — Smart India Hackathon 2026 (Internal Round) · Frontend Developer · Team Shark

— Participated — Radiothon Winter 2026 Hackathon · ML Engineer & Lead · Team Doom

— Active member, Data Science Club at VIT-AP University · Documentation Team

— Open-source projects on GitHub (GokulM8)